Final Cheatsheet — Lectures 3–4

Definitions and theorems that are easy to forget. Vectors are columns; log is the natural logarithm.

Definitions people forget

Derivatives

- **Jacobian** of $f:\mathbb{R}^n\to\mathbb{R}^m$: $m\times n$, $(Df)_{ij}=\frac{\partial f_i}{\partial x_j}$; rows are $\nabla f_i^\top$. $Df=\nabla f^\top$ for scalar f.
- **Differentiable**: $f(\mathbf{x}+\mathbf{h})=f(\mathbf{x})+Df(\mathbf{x})\mathbf{h}+o(\|\mathbf{h}\|)$. Directional derivative $D_{\mathbf{v}}f=\nabla f^\top\mathbf{v}$.
- **Gradient w.r.t. a matrix**: same shape as X, $(\nabla f(X))_{ij}=\frac{\partial f}{\partial X_{ij}}$. The derivative of $F:\mathbb{R}^{m\times n}\to\mathbb{R}^{p\times q}$ is the order-4 tensor $\frac{\partial F_{ij}}{\partial X_{kl}}$.
- **Hessian** $(\nabla^2 f)_{ij}=\frac{\partial^2 f}{\partial x_i\partial x_j}$, symmetric for C^2 functions (Schwarz).

Optimization

- **L-smooth**: $\|\nabla f(\mathbf{x})-\nabla f(\mathbf{y})\|\le L\|\mathbf{x}-\mathbf{y}\|$. For C^2: $\iff\|\nabla^2 f(\mathbf{x})\|_2\le L$ for all $\mathbf{x}$.
- **Convex**: $f(\theta\mathbf{x}+(1-\theta)\mathbf{y})\le\theta f(\mathbf{x})+(1-\theta)f(\mathbf{y})$.
- **μ-strongly convex**: $f(\mathbf{y})\ge f(\mathbf{x})+\nabla f(\mathbf{x})^\top(\mathbf{y}-\mathbf{x})+\frac{\mu}{2}\|\mathbf{y}-\mathbf{x}\|^2$. For C^2: $\iff$ all eigenvalues of all Hessians $\ge\mu$. Then there is exactly one minimizer.
- **Subgradient** $\mathbf{g}\in\partial f(\mathbf{x})$: $f(\mathbf{y})\ge f(\mathbf{x})+\mathbf{g}^\top(\mathbf{y}-\mathbf{x})$ for all $\mathbf{y}$. $\partial|x|(0)=[-1,1]$.
- **Prox**: $\operatorname{prox}_{\eta h}(\mathbf{v})=\arg\min_{\mathbf{u}}\left(h(\mathbf{u})+\frac{1}{2\eta}\|\mathbf{u}-\mathbf{v}\|^2\right)$; for $h=\lambda\|\cdot\|_1$: soft thresholding $S_{\eta\lambda}(v)=\operatorname{sign}(v)\max(|v|-\eta\lambda,0)$.
- **Lagrangian** $\mathcal{L}(\mathbf{x},\boldsymbol{\mu})=f+\sum_i\mu_i g_i$ for constraints $g_i\le 0$; **dual function** $d(\boldsymbol{\mu})=\inf_{\mathbf{x}}\mathcal{L}(\mathbf{x},\boldsymbol{\mu})$, $\boldsymbol{\mu}\ge\mathbf{0}$.
- **Convex conjugate** $f^*(\mathbf{y})=\sup_{\mathbf{x}}(\mathbf{y}^\top\mathbf{x}-f(\mathbf{x}))$.

Probability

- **Quantile function** $F^{-1}(u)=\min\{x:F(x)\ge u\}$.
- $\operatorname{Cov}(X,Y)=\mathbb{E}[XY]-\mathbb{E}X\,\mathbb{E}Y$; $\operatorname{Corr}=\frac{\operatorname{Cov}}{\sigma_X\sigma_Y}$. Uncorrelated $\not\Rightarrow$ independent (except jointly Gaussian).
- **Conditional independence** $X\perp\!\!\!\perp Y\mid Z$: $p(x,y\mid z)=p(x\mid z)p(y\mid z)$. Chain $X\to Z\to Y$ and fork: $X\perp\!\!\!\perp Y\mid Z$. Collider $X\to Z\leftarrow Y$: $X\perp\!\!\!\perp Y$, but in general not given Z.
- **MGF** $M_X(t)=\mathbb{E}[e^{tX}]$, $K_X=\log M_X$; $K_X'(0)=\mathbb{E}X$, $K_X''(0)=\operatorname{Var}X$.
- $\Sigma=\operatorname{Cov}(\mathbf{X})=\mathbb{E}[(\mathbf{X}-\boldsymbol{\mu})(\mathbf{X}-\boldsymbol{\mu})^\top]$. **Mahalanobis**: $\sqrt{(\mathbf{x}-\boldsymbol{\mu})^\top\Sigma^{-1}(\mathbf{x}-\boldsymbol{\mu})}$; whitening $\Sigma^{-1/2}(\mathbf{X}-\boldsymbol{\mu})$.
- **In probability**: $P(|Y_n-Y|>\varepsilon)\to 0$ for all $\varepsilon>0$. **In distribution**: $P(Y_n\le y)\to P(Y\le y)$ where the CDF of Y is continuous.

Statistics

- **Bias** $\mathbb{E}\hat\theta-\theta$; **MSE** $\mathbb{E}[(\hat\theta-\theta)^2]$; **consistent**: $\hat\theta_N\to\theta$ in probability.
- **Score** $s=\frac{\partial}{\partial\theta}\log p(x\mid\theta)$, $\mathbb{E}_\theta s=0$. **Fisher information** $I(\theta)=\mathbb{E}[s^2]=-\mathbb{E}\left[\frac{\partial^2}{\partial\theta^2}\log p(X\mid\theta)\right]$ (one observation).
- **Sufficient** T: $p(\mathbf{x}\mid\theta)=g(T(\mathbf{x}),\theta)h(\mathbf{x})$.
- **Exponential family** $p_{\boldsymbol{\eta}}(x)=h(x)\exp(\boldsymbol{\eta}^\top T(x)-A(\boldsymbol{\eta}))$.
- **Beta**(a,b): $\propto\theta^{a-1}(1-\theta)^{b-1}$, mean $\frac{a}{a+b}$, mode $\frac{a-1}{a+b-2}$.
- **GP** $f\sim\mathcal{GP}(m,k)$: every $(f(\mathbf{x}_1),\dots,f(\mathbf{x}_n))$ is Gaussian with $K_{ij}=k(\mathbf{x}_i,\mathbf{x}_j)$.
- $H(p)=-\sum p_k\log p_k$, $H(p,q)=-\sum p_k\log q_k$, $\operatorname{KL}(p\,\|\,q)=\sum p_k\log\frac{p_k}{q_k}=H(p,q)-H(p)$.

Distributions

	$\mathbb{E}X$	$\operatorname{Var}X$
Bernoulli(p)	p	$p(1-p)$
Binomial(n,p)	np	$np(1-p)$
Geometric(p) on $\{1,2,\dots\}$	$\frac{1}{p}$	$\frac{1-p}{p^2}$
Poisson(λ), $e^{-\lambda}\frac{\lambda^k}{k!}$	λ	λ
Uniform(a,b)	$\frac{a+b}{2}$	$\frac{(b-a)^2}{12}$
Exponential(λ), $\lambda e^{-\lambda x}$	$\frac{1}{\lambda}$	$\frac{1}{\lambda^2}$

$\mathcal{N}(\boldsymbol{\mu},\Sigma)$: $p(\mathbf{x})=\frac{1}{(2\pi)^{d/2}\det(\Sigma)^{1/2}}\exp\left(-\frac{1}{2}(\mathbf{x}-\boldsymbol{\mu})^\top\Sigma^{-1}(\mathbf{x}-\boldsymbol{\mu})\right)$.

Theorems people forget

Derivatives

- **Chain rule** $D(f\circ g)(\mathbf{x})=Df(g(\mathbf{x}))\,Dg(\mathbf{x})$; scalar f: $\nabla(f\circ g)=Dg^\top\nabla f(g)$. Linear layer $\mathbf{y}=W\mathbf{x}+\mathbf{b}$: $\nabla_W L=\boldsymbol{\delta}\mathbf{x}^\top$, $\nabla_{\mathbf{x}}L=W^\top\boldsymbol{\delta}$.
- $\nabla(\mathbf{a}^\top\mathbf{x})=\mathbf{a}$; $\nabla(\mathbf{x}^\top A\mathbf{x})=(A+A^\top)\mathbf{x}$; $\nabla\|\mathbf{x}\|=\frac{\mathbf{x}}{\|\mathbf{x}\|}$; $\nabla\|A\mathbf{x}-\mathbf{b}\|^2=2A^\top(A\mathbf{x}-\mathbf{b})$.
- $\nabla_X\operatorname{tr}(AX)=A^\top$; $\nabla_X(\mathbf{a}^\top X\mathbf{b})=\mathbf{a}\mathbf{b}^\top$; $\nabla_X\|AX-B\|_F^2=2A^\top(AX-B)$; $\nabla_X\operatorname{tr}(X^\top AX)=(A+A^\top)X$.
- **Jacobi**: $\nabla_X\det X=\det(X)X^{-\top}$, $\nabla_X\log|\det X|=X^{-\top}$. Inverse: $(X+H)^{-1}=X^{-1}-X^{-1}HX^{-1}+o(\|H\|)$, so $\nabla_X\operatorname{tr}(AX^{-1})=-X^{-\top}A^\top X^{-\top}$.
- Softmax/sigmoid: $\sigma'=\sigma(1-\sigma)$; $\nabla\log\sum_j e^{z_j}=\operatorname{softmax}(\mathbf{z})$, Hessian $\operatorname{diag}(\mathbf{p})-\mathbf{p}\mathbf{p}^\top$.
- **Taylor**: $f(\mathbf{x}+\mathbf{h})=f(\mathbf{x})+\nabla f(\mathbf{x})^\top\mathbf{h}+\frac{1}{2}\mathbf{h}^\top\nabla^2 f(\mathbf{x}+\tau\mathbf{h})\mathbf{h}$, $\tau\in(0,1)$.

Optimization

- **Second-order test** at $\nabla f=\mathbf{0}$: Hessian eigenvalues all >0: strict local min; all <0: max; both signs: saddle.
- **Descent lemma** (L-smooth): $f(\mathbf{y})\le f(\mathbf{x})+\nabla f(\mathbf{x})^\top(\mathbf{y}-\mathbf{x})+\frac{L}{2}\|\mathbf{y}-\mathbf{x}\|^2$. GD with $\eta\le\frac{1}{L}$: $\min_{k<K}\|\nabla f(\mathbf{x}_k)\|^2\le\frac{2(f(\mathbf{x}_0)-f^*)}{\eta K}$. Armijo: $f(\mathbf{x}-\eta\nabla f)\le f(\mathbf{x})-\frac{\eta}{2}\|\nabla f\|^2$.
- **Quadratics** (A SPD): GD converges $\iff 0<\eta<\frac{2}{\lambda_{\max}}$; best $\eta=\frac{2}{\lambda_{\max}+\lambda_{\min}}$, rate $\frac{\kappa-1}{\kappa+1}$.
- **Rates**, convex L-smooth, $\eta=\frac{1}{L}$: $f(\mathbf{x}_k)-f^*\le\frac{L\|\mathbf{x}_0-\mathbf{x}^*\|^2}{2k}$; if also μ-strongly convex: $\le(1-\frac{\mu}{L})^k(f(\mathbf{x}_0)-f^*)$. PL: $f-f^*\le\frac{1}{2\mu}\|\nabla f\|^2$.
- **Newton** $\mathbf{x}_{k+1}=\mathbf{x}_k-\nabla^2 f(\mathbf{x}_k)^{-1}\nabla f(\mathbf{x}_k)$: quadratic convergence near a minimum with a PD, Lipschitz Hessian.
- Convex: $f(\mathbf{y})\ge f(\mathbf{x})+\nabla f(\mathbf{x})^\top(\mathbf{y}-\mathbf{x})\iff\nabla^2 f\succeq 0$; local min = global min. $\mathbf{x}^*$ minimizes $f+h\iff-\nabla f(\mathbf{x}^*)\in\partial h(\mathbf{x}^*)$.
- **KKT** ($g_i\le 0$, LICQ): $\nabla f+\sum\mu_i\nabla g_i=\mathbf{0}$; $g_i\le 0$; $\mu_i\ge 0$; $\mu_i g_i=0$. Convex f, g_i: KKT $\Rightarrow$ global min. Equality constraints: λ_i of any sign.
- **Weak duality** $d(\boldsymbol{\mu})\le f(\mathbf{x})$ for feasible $\mathbf{x}$; convex + KKT point: $d^*=p^*$. **Fenchel–Young**: $f(\mathbf{x})+f^*(\mathbf{y})\ge\mathbf{x}^\top\mathbf{y}$.

Probability

- **Bayes**: $P(B_j\mid A)=\frac{P(A|B_j)P(B_j)}{\sum_i P(A|B_i)P(B_i)}$.
- **Change of variables**: $p_Y(y)=p_X(x)/|g'(x)|$, $x=g^{-1}(y)$. $F^{-1}(U)\sim F$ for $U\sim\text{Unif}(0,1)$.
- **Tail sums**: $\mathbb{E}X=\sum_{k\ge0}P(X>k)$ on $\{0,1,\dots\}$; $\mathbb{E}X=\int_0^\infty P(X>t)\,dt$ for $X\ge 0$.
- **Markov** $P(X\ge a)\le\frac{\mathbb{E}X}{a}$; **Chebyshev** $P(|X-\mu|\ge k)\le\frac{\sigma^2}{k^2}$; **Chernoff** $P(X\ge a)\le\inf_{t>0}e^{-ta}M_X(t)$; **Hoeffding** (i.i.d., values in $[a,b]$) $P(|\bar X_N-\mu|\ge\varepsilon)\le 2e^{-2N\varepsilon^2/(b-a)^2}$; **Jensen** $f(\mathbb{E}X)\le\mathbb{E}f(X)$ for convex f.
- $\operatorname{Var}(X+Y)=\operatorname{Var}X+\operatorname{Var}Y+2\operatorname{Cov}(X,Y)$; $\operatorname{Cov}(A\mathbf{X}+\mathbf{b})=A\Sigma A^\top$; $\operatorname{Var}(\mathbf{u}^\top\mathbf{X})=\mathbf{u}^\top\Sigma\mathbf{u}$. Covariance is a tensor: $\operatorname{Cov}(A\mathbf{X})_{ab}=A_{ai}A_{bj}\Sigma_{ij}$.
- **Tower** $\mathbb{E}[\mathbb{E}[Y\mid X]]=\mathbb{E}Y$; $\mathbb{E}[Y\mid X]$ is the best predictor in mean square. **Total variance**: $\operatorname{Var}Y=\mathbb{E}[\operatorname{Var}(Y\mid X)]+\operatorname{Var}(\mathbb{E}[Y\mid X])$.
- MGF: $M_{aX+b}(t)=e^{bt}M_X(at)$; independent: $M_{X+Y}=M_XM_Y$.
- **Gaussians**: $A\mathbf{Z}+\boldsymbol{\mu}\sim\mathcal{N}(\boldsymbol{\mu},AA^\top)$. Conditional: $\mathbf{X}_a\mid\mathbf{x}_b\sim\mathcal{N}\big(\boldsymbol{\mu}_a+\Sigma_{ab}\Sigma_{bb}^{-1}(\mathbf{x}_b-\boldsymbol{\mu}_b),\ \Sigma_{aa}-\Sigma_{ab}\Sigma_{bb}^{-1}\Sigma_{ba}\big)$. Block-diagonal $\Sigma\Rightarrow$ independent blocks.
- $\operatorname{Var}\bar X_N=\frac{\sigma^2}{N}$. **WLLN**: $P(|\bar X_N-\mu|\ge\varepsilon)\le\frac{\sigma^2}{N\varepsilon^2}$. **CLT**: $\frac{\sqrt{N}(\bar X_N-\mu)}{\sigma}\to\mathcal{N}(0,1)$ in distribution.
- **Slutsky**: $X_n\to X$ in dist., $Y_n\to c$ in prob. $\Rightarrow X_n+Y_n\to X+c$, $X_nY_n\to cX$. **Delta method**: $\sqrt{N}(T_N-\theta)\to\mathcal{N}(0,\tau^2)\Rightarrow\sqrt{N}(g(T_N)-g(\theta))\to\mathcal{N}(0,g'(\theta)^2\tau^2)$.
- **Rejection sampling**: accept w.p. $\frac{p(X)}{Mq(X)}$, overall rate $\frac{1}{M}$. **Importance sampling**: $\mathbb{E}_p f=\mathbb{E}_q[f\frac{p}{q}]$.

Statistics and information

- MSE = bias^2+Var. $s^2=\frac{1}{N-1}\sum(X_i-\bar X)^2$ is unbiased; the Gaussian MLE uses $\frac{1}{N}$.

- **Cramér–Rao**: unbiased $\Rightarrow \operatorname{Var}\hat\theta \geq \frac{1}{NI(\theta)}$. MLE: $\sqrt{N}(\hat\theta-\theta) \to \mathcal{N}(0, \frac{1}{I(\theta)})$; SE $\approx (-\ell''(\hat\theta))^{-1/2}$.
- **Rao–Blackwell**: $\mathbb{E}[\hat\theta \mid T]$, T sufficient, has the same bias and no larger MSE.
- Exp. family: $\nabla A(\boldsymbol{\eta}) = \mathbb{E}[T(X)]$, $\nabla^2 A = \operatorname{Cov}(T(X)) \succeq 0$; MLE solves $\nabla A(\hat{\boldsymbol{\eta}}) = \frac{1}{N}\sum_i T(x_i)$.
- Least squares = Gaussian MLE; ridge = MAP with $\mathbf{w} \sim \mathcal{N}(\mathbf{0}, \alpha^{-1}I)$, $\lambda = \alpha\sigma^2$.
- **Beta–Bernoulli**: $\text{Beta}(a,b) + k$ ones in $N \to \text{Beta}(a+k, b+N-k)$.
- **Bayesian linear regression**: $S = (\alpha I + \sigma^{-2}X^\top X)^{-1}$, $\mathbf{m} = \sigma^{-2}SX^\top\mathbf{y}$; predictive $\mathcal{N}(\mathbf{m}^\top\mathbf{x}_*, \mathbf{x}_*^\top S\mathbf{x}_* + \sigma^2)$.
- **GP regression**: mean $\mathbf{k}_*^\top(K+\sigma^2 I)^{-1}\mathbf{y}$, variance $k(\mathbf{x}_*,\mathbf{x}_*) - \mathbf{k}_*^\top(K+\sigma^2 I)^{-1}\mathbf{k}_*$.
- **Gibbs**: $\mathrm{KL}(p\,\|\,q) \geq 0, = 0 \iff p = q$. MLE minimizes $\mathrm{KL}(\hat p\,\|\,q_\theta) =$ cross-entropy. $\mathrm{KL}(\mathcal{N}(\mu_1,\sigma_1^2)\,\|\,\mathcal{N}(\mu_2,\sigma_2^2)) = \log\frac{\sigma_2}{\sigma_1} + \frac{\sigma_1^2+(\mu_1-\mu_2)^2}{2\sigma_2^2} - \frac{1}{2}$.
- **ELBO**: $\log p(\mathbf{x}\mid\theta) = \sum_z q(z)\log\frac{p(\mathbf{x},z\mid\theta)}{q(z)} + \mathrm{KL}(q\,\|\,p(\cdot\mid\mathbf{x},\theta))$. EM: E-step q = posterior, M-step maximizes $\sum_i\sum_z q_i(z)\log p(\mathbf{x}_i, z\mid\theta)$; never decreases ℓ. GMM: $\pi_k = \frac{N_k}{N}$, $\boldsymbol{\mu}_k = \frac{1}{N_k}\sum_i r_{ik}\mathbf{x}_i$.